

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 12

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Vehicles: high energy density (200 Wh kg^{-1}) minimises battery mass so range is greater; many cycles and high efficiency (1) • Stationary storage: mass is unimportant so the low energy density of lead-acid does not matter; lowest cost per kWh; well-established, recyclable (1) • Lead-acid drawbacks: fewer cycles and lower efficiency; lithium fire risk needs management (1) • Nickel-cadmium largely replaced because of cadmium toxicity and cost (1)	4
01.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Recharging speed (1) • Self-discharge during non-use (1) • Availability of raw materials (lithium, cobalt) (1) • Recyclability / disposal (1)	2
01.3	<i>1 mark for each point.</i> • Parked electric vehicles plugged into the grid form a very large distributed battery (1) • At peaks a small proportion of their stored energy is fed back to the grid; surplus grid energy recharges them - avoiding costly, low-efficiency open-cycle gas turbines (1)	2
01.4	<i>1 mark for each point.</i> • Surplus electricity electrolyses water to hydrogen, which can be converted to methane (1) • Methane is fed into the existing natural gas pipe network and storage, so no new distribution system is needed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 223-224; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • 1 nitrogen fixation; 2 nitrification (1) • 3 denitrification; 4 ammonification (1) • 5 ionisation; 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three) (1)	3
02.2	<i>1 mark for each point.</i> • Drainage makes the soil aerobic (1) • Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate (1) • Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas (1)	3
02.3	<i>1 mark for each point.</i> • Nitrogen gas is very unreactive so most organisms cannot use it directly (1) • Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate; the cycle recycles a small usable amount (1)	2
02.4	<i>1 mark for each point.</i> • Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins (1) • When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	
03.1	<i>1 mark for each point.</i> • There is an environmental gradient (distance from the hedge / changing conditions) so samples must be taken at regular intervals along it to detect the change (1) • Random sampling might leave gaps / miss the pattern; systematic sampling avoids bias while covering the gradient (1)	2
03.2	<i>1 mark for each point.</i> • Sample at close intervals (eg every 1 m) first and see at what spacing the variations are still detected without unnecessary data; wider intervals (eg 4 m) may miss peaks and troughs (1) • Take increasing numbers of samples at one site and calculate the running mean; the number needed is where the mean stops fluctuating / stabilises (1) • Calculate standard deviation to assess scatter: more variable data need more samples (1) • Test the method (extraction technique, timing) for reliability before the main study (1)	4
03.3	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Remove surface vegetation; pour an irritant solution (dilute mustard / detergent in water) onto the area until saturated (and around it), keep adding as it infiltrates (up to 2) • Collect worms that come to the surface for a standardised time; count / identify / weigh (up to 2) • Limitations: solution may not percolate impermeable soils or reach deep worms; worms may move sideways or downwards; some may die or not respond; slow-moving worms missed (up to 2)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
04.1	<i>1 mark for each point.</i> - Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended (1) - Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges (1) - The particles clump together (flocs) and settle in the clarifier tank (1)	3
04.2	<i>1 mark for each point.</i> - Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources (1) - Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed; improves taste (1)	2
04.3	<i>1 mark for each point.</i> - Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) (1) - Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution) (1)	2
04.4	<i>1 mark for each point.</i> - Ion exchange: toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) (1) - Reverse osmosis / distillation (desalination) - expensive, energy-intensive (1) - Activated carbon: organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	
05.1	<i>1 mark.</i> The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) (1)	1
05.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> - Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions (1) - Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition (1) - Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils (1) - pH / presence of other chemicals may increase or decrease the rate of degradation (1) - Adsorption onto clay / organic matter may protect from degradation and increase persistence (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
06.1	<i>1 mark.</i> Temperature inversion (1)	1

Question	Marking guidance	Marks
06.2	<i>1 mark for each point.</i> • On night A, temperature falls with altitude; warm effluent gases are less dense / more buoyant than the surrounding air so they rise (1) • As they rise they cool but the surrounding air is also cooler so they stay buoyant and continue to rise, dispersing and diluting (1) • On night B the air near the ground is colder / denser than the air above (inversion layer at about 400 m) (1) • Pollutants cool as they rise and become denser than the warmer air above so they are trapped below the inversion layer and become concentrated (1)	4
06.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Cloudless sky allows infrared energy to be radiated from the ground so the ground cools rapidly (1) • The cold ground cools the air in contact with it, producing a cold dense layer near the ground (1) • Low wind velocity means the air layers of different temperatures do not mix (1) • Valley allows cold dense air to collect / drain into the low ground (1)	3
06.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Smoke smogs (eg London 1952) (1) • Photochemical smogs (eg Los Angeles, Mexico City) (1) • Build-up of NO_x / carbon monoxide / PM₁₀ in cities (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 241-242; AQA spec 3.4.2.2.	
07.1	<i>1 mark for each point.</i> • Gasification: coal (including deposits too deep to mine) is burnt underground under controlled conditions to produce fuel gases (hydrogen, carbon monoxide, methane) (1) • Allows deep coal to be exploited and gas can be piped / used in gas turbines or for hydrogen with CCS (1) • Liquefaction: coal is converted to liquid hydrocarbons directly with solvents or via gasification and chemical conversion (1) • Provides liquid fuels for vehicles / aircraft that solid coal cannot supply (1)	4
07.2	<i>1 mark for each point.</i> • A solid, ice-like crystalline structure in which methane is trapped within water ice (1) • Found at low temperatures (polar regions / permafrost) or high pressure in ocean sediments around continental margins (1)	2
07.3	<i>1 mark for each point.</i> • Water heating: hot water pumped into sediments melts the hydrate releasing methane (1) • Depressurisation: drilling reduces pressure so methane dissociates from the crystals (1) • CO₂ injection: CO₂ bonds more strongly to the ice and displaces methane (could double as CCS) (1) • Risks: methane not collected escapes to the atmosphere - powerful greenhouse gas / positive feedback; seabed ecology poorly understood; sediment instability / landslides (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187, 190; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
08.1	<i>1 mark.</i> As the (linear) purity / grade of a mineral deposit decreases, the amount of the mineral present increases exponentially / logarithmically (1)	1
08.2	<i>1 mark for the correct answer.</i> $2400 / 40 = 60$ times (1)	1
08.3	<i>1 mark for each point.</i> - Reserves: the amount that can be exploited now, economically, with existing technology (deposits above the cut-off ore grade) (1) - Resource: all the material that could be exploited technically and economically now or in the future (with realistic price rises or new technology), including deep, low-grade or protected deposits (1) - The resource is finite but the reserves change: they rise if prices rise or technology improves, and fall if prices fall (1)	3
08.4	<i>1 mark for each point.</i> - Lowering the cut-off ore grade includes deposits of lower purity in the reserves (1) - Because the amount of material rises exponentially as purity falls, a small drop in COOG adds a very large mass (eg 0.5-1.0% has 4x the copper of 1.0-2.0%) (1)	2
08.5	<i>1 mark for each point.</i> - The total quantity (stock / resource base) is far greater than could ever be used (1) - But most is at purities too low or in locations too inaccessible for current economic exploitation (1) - The challenge is developing technologies (low-energy extraction such as bioleaching, phytomining; better exploration) that make low-grade deposits exploitable; energy needs rise rapidly as ore grade falls (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 135-136; AQA spec 3.2.3.3.	
09.1	<i>1 mark for each point.</i> - pH 5.6 (1) - Atmospheric carbon dioxide dissolves in rainwater to form (weak) carbonic acid (1)	2
09.2	<i>1 mark for each point.</i> - Limestone bedrock (calcium carbonate) is alkaline / basic and neutralises the acids as water percolates through it, so the lake pH stays high (7.2) (1) - Granite does not neutralise acids so lake water in A is acidic (4.6) (1) - Low pH mobilises / dissolves aluminium ions from the soil and rock (0.42 mg dm^{-3} in A) which are toxic to fish (damage gills) (1) - Acids also damage fish eggs / gills (exposed living cells) and dissolve the exoskeletons of invertebrate food; lichens and other food organisms lost (1)	4
09.3	<i>1 mark for each point.</i> - Corrodes metals: railway lines, railings, pipes, pylons, overhead power lines (1) - Dissolves / weakens limestone and marble buildings and statues (eg York Minster) (1)	2

Question	Marking guidance	Marks
09.4	<i>1 mark for each point.</i> • Sulfur dioxide is oxidised (by ozone / other oxidants) to sulfur trioxide (1) • Sulfur trioxide dissolves in water droplets to form sulfuric acid (SO₂ alone gives the weaker sulfurous acid) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 250-252; AQA spec 3.4.3.2.2.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Properties of renewables: intermittency, predictability, low energy density, storage, applicability, locational constraints, impacts, technology level, economics • Wind: large offshore resource in shallow seas, falling costs, HAWT / VAWT / direct drive developments; intermittent, needs storage and grid; land-use conflicts • Solar: PV costs falling rapidly, roof space, but low winter insolation in the UK; CSP unsuitable; heat pumps and photothermal for heating • HEP: most large sites used; low-head, helical and micro-hydro extend use; pumped storage vital • Tidal: large range (Severn, Mersey), predictable; barrages costly with high impact, lagoons and in-stream turbines lower impact • Wave: long Atlantic fetch, but technology immature • Biofuels: wastes, anaerobic digestion, energy crops vs food / habitat • Geothermal: limited to low-temperature schemes / heat pumps • Systems issues: storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management, diversity of sources, energy conservation reducing demand, electrification of transport • Judgement: renewables could supply most electricity with storage and a diverse mix, but heat and transport are harder; transition needs investment and lifestyle change	9
10.2	<i>1 mark.</i> • Eight times (energy proportional to velocity cubed) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-217; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
11.1	Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why non-renewable: formed too slowly; geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock; Lasky's principle; cut-off ore grade and its drivers • Extraction technologies: mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • Reducing demand: recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • Environmental limits: impacts of mining and their control; energy costs rising with lower grades; conflicts • Judgement: technology can extend supplies substantially but only within energy and environmental constraints; circular economy essential Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility; DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • Reducing impact: restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • Alternatives: cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • Trade-offs: yield, cost, human health (organophosphates), resistance evolution, pollinator services; risk:benefit; sustainability and the circular economy • Judgement: no pesticide is perfectly specific; combination of property-based selection and integrated control is most sustainable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	25